Stable Diffusion

Naeemullah Khan

naeemullah.khan@kaust.edu.sa

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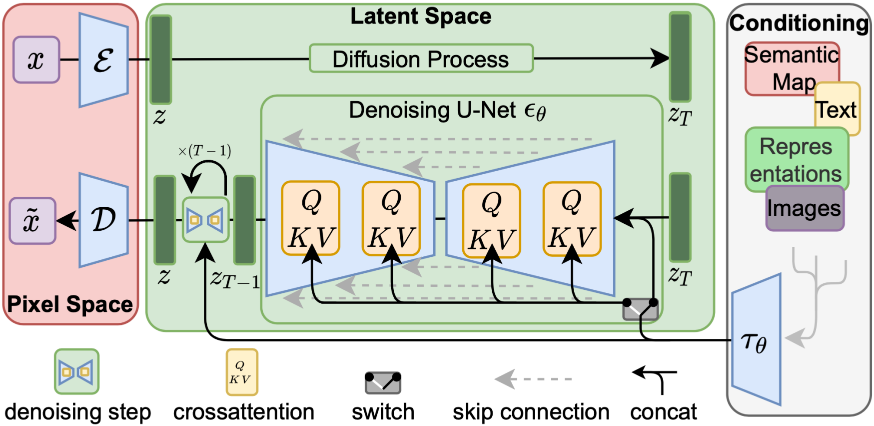
Latent Space
Pixel Space
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Diffusion Process
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z
z_{T-1}
Q
KV
Semantic Map
Text
Repres entations
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τ_θ
denoising step
crossattention
switch
skip connection
concat

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What is Stable Diffusion?

- A state-of-the-art text-to-image generative model.
- Developed by Stability AI and collaborators (2022).
- Open-source, enabling wide adoption and customization.

Advantages

- High-quality, diverse image generation.
- Efficient training and inference due to latent space operations.
- Open-source and extensible for research and applications.

What do we need to make a good generative model?

1. Method of learning to generate new images — **Forward/reverse diffusion**
2. Way to link text and images — **Text-image representation model**
3. Way to compress images — **Autoencoder**
4. Way to add in good inductive biases — **U-net architecture + 'attention'**

Making a 'good' generative model is about making all these parts work together well!

The Journey of Generative Modeling:

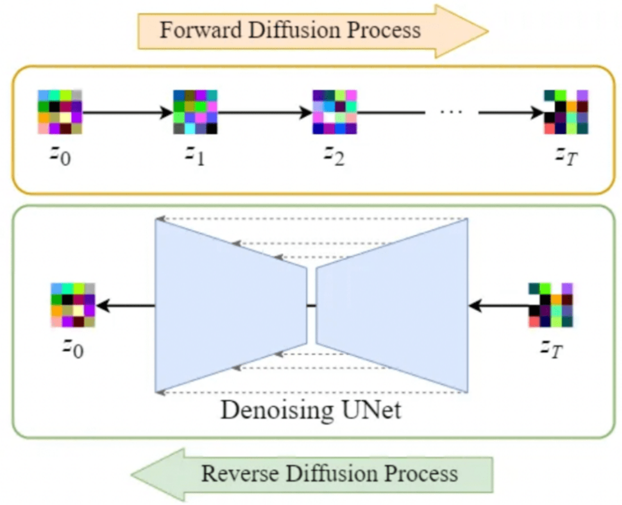
- Early successes: GANs and VAEs enabled impressive image synthesis and data generation.
- Drawbacks:
  - GANs: Training instability, mode collapse, and sensitivity to hyperparameters.
  - VAEs: Blurry outputs and limited expressiveness.
- Key Challenge: Achieving both high sample quality and stable, reliable training.

Where do we go from here?

- Need for models that are robust, interpretable, and capable of generating diverse, high-fidelity samples.

Enter Diffusion Models:

- Inspired by thermodynamics, diffusion models gradually transform noise into data.
- Highlights:
  - Simple training objective
  - Stable optimization
  - State-of-the-art sample quality
  - Probabilistic and interpretable framework


Forward Diffusion Process
z_0
z_1
z_2
z_T
z_0
z_T
Denoising UNet
Reverse Diffusion Process

Diffusion Models: **Introduction**

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- **Inspiration:** Rooted in concepts from thermodynamics and stochastic processes.
- **Key Idea:**
  - *Forward process:* Progressively corrupt data by adding Gaussian noise.
  - *Reverse process:* Learn to recover the original data by reversing the noising process.

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- **Inspiration:** Rooted in concepts from thermodynamics and stochastic processes.
- **Key Idea:**
  - *Forward process:* Progressively corrupt data by adding Gaussian noise.
  - *Reverse process:* Learn to recover the original data by reversing the noising process.
- **Popularity:** Known for producing high-quality and diverse generated samples.

Quick Overview

- **Forward:** Clean $\rightarrow$ Noise (additive Gaussian steps)
- **Reverse:** Learn **U-Net** to remove noise step by step
- **Training:** Model predicts the noise added
- **Sampling:** Roll back from pure noise to a data sample

Diffusion Models: **Denoising**

Denoising Diffusion Probabilistic Models

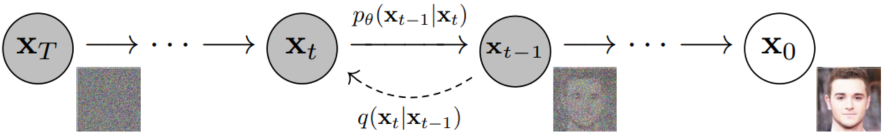
Denoising diffusion models, also known as score-based generative models, have recently emerged as a powerful class of generative models. They achieve remarkable results in high-fidelity image generation, often outperforming GANs.

Diffusion Models Beat GANs on Image Synthesis Dhariwal & Nichol, OpenAI, 2021

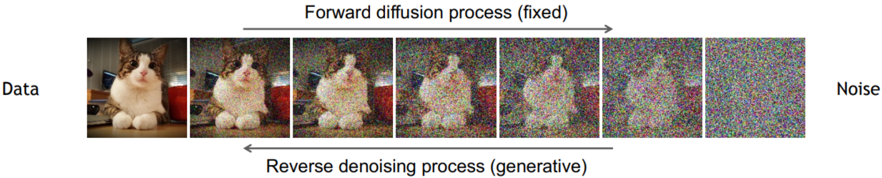
Denoising Diffusion Probabilistic Models (cont.)

Denoising diffusion models consist of two processes:

- A fixed (predefined) **forward diffusion process** q, which gradually adds Gaussian noise to an image until only pure noise remains.
- A learned **reverse denoising diffusion process** p_θ, where a neural network is trained to gradually denoise an image starting from pure noise, eventually recovering the original image.



Denoising Diffusion Probabilistic Models (cont.)



Diffusion Models: **Forward Process**

Forward Diffusion Process

- The forward process is a Markov chain that iteratively adds Gaussian noise to the image at each timestep.
- Given clean data $\mathbf{x}_0$, we generate $\mathbf{x}_t$ as:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

- Over many steps (e.g., 1000), the data becomes indistinguishable from pure noise.

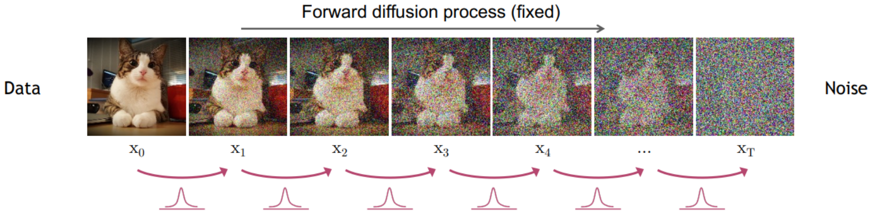
Forward Diffusion Process (cont.)

- This process is predefined and not learned.
- The full forward process:

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1})$$

- Each new (slightly noisier) image at time step t is drawn from a conditional Gaussian distribution with $\mu_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1}$ and $\sigma_t^2 = \beta_t$, which we can do by sampling $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and then setting $\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\epsilon$.

Forward Diffusion Process (cont.)



- What if we want to go directly from q_0 to q_t?
- Let $\overline{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$, then

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\overline{\alpha}_t}\mathbf{x}_0, (1-\overline{\alpha}_t)\mathbf{I})$$

- This allows us, during training, to optimize random terms of the loss function L (i.e., randomly sample t and optimize L_t).

Diffusion Models: **Reverse Denoising Process**

Reverse Denoising Process

In the reverse denoising process, we denoise Gaussian noise to generate an image.

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- At each time step t, we want to sample $\mathbf{x}_{t-1}$ given $\mathbf{x}_t$.
- The reverse process is defined as $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$, which we will learn using a neural network.

Reverse Denoising Process

- The reverse process is a Markov chain that iteratively denoises the image.
- We want to sample $\mathbf{x}_{t-1}$ from $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$.
- However, $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is unknown and needs to be learned.

Reverse Denoising Process (cont.)

- We approximate it with a neural network that predicts the mean and variance of the Gaussian distribution.
- If β_t is small enough at each time step, we can assume $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is a Gaussian distribution:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t))$$

where μ_θ and Σ_θ are neural networks conditioned on $\mathbf{x}_t$ and t.

Reverse Denoising Process (cont.)

- For simplicity, we can fix the variance Σ_θ to a constant value (e.g., $\beta_t \mathbf{I}$) and only learn the mean μ_θ.

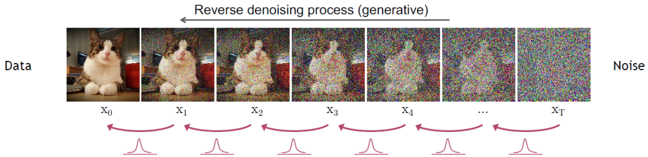
Reverse Denoising Process (cont.)

- The reverse process can then be expressed as:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \beta_t \mathbf{I})$$

and the joint distribution over all time steps is:

$$p_\theta(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$$



Reverse denoising process in diffusion models.

Diffusion Models: **Training & Sampling Algorithms**

Algorithm

Algorithm 1 Training

1: **repeat**
2: $\quad \mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $\quad t \sim \mathrm{Uniform}(\{1, \ldots, T\})$
4: $\quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: $\quad$ Take gradient descent step on

$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\left(\boxed{\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}}, t\right) \right\|^2$$

6: **until** converged

Training Algorithm

1. Sample x_0 from the data distribution.
2. Randomly select a timestep t.
3. Corrupt x_0 with noise to obtain x_t.
4. Model predicts noise $\epsilon_\theta(x_t, t)$.
5. Compute loss: mean squared error between predicted and true noise.
6. Optimize using SGD or Adam.

Algorithm (cont.)

Algorithm 2 Sampling

$\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for $t = T, \ldots, 1$ **do**
  $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
  $\mathbf{x}_{t-1} = \boxed{\frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right)} + \sigma_t \mathbf{z}$
end for
return $\mathbf{x}_0$

Sampling Algorithm

1. Start from Gaussian noise x_T.
2. For each timestep $t = T, \ldots, 1$:
   - 2.1 Predict noise $\epsilon_\theta(x_t, t)$.
   - 2.2 Compute posterior mean $\mu_\theta(x_t, t)$.
   - 2.3 Sample x_{t-1} from the Gaussian posterior.
   - 2.4 (Optional) Apply guidance (e.g., classifier-free) for improved results.

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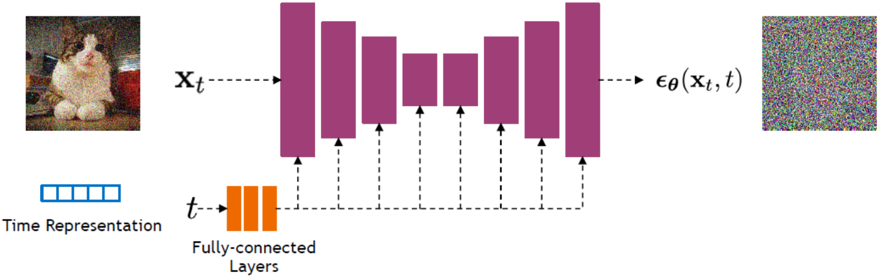
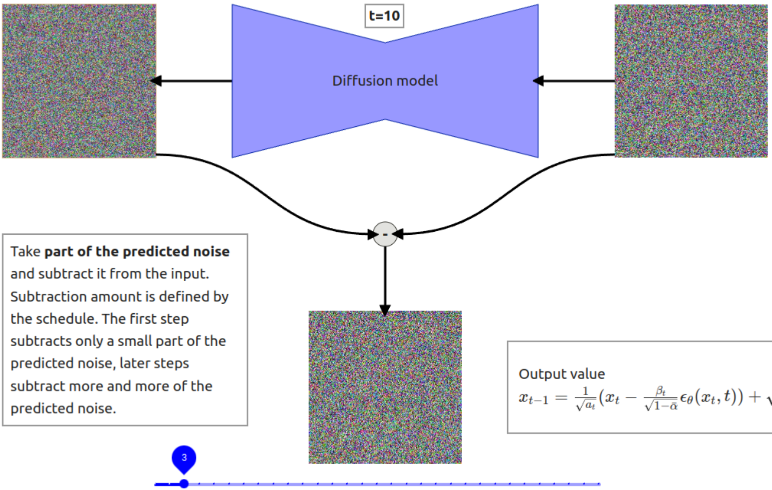
Diffusion Models: **Network Architectures**

Network Architectures

- Diffusion models commonly use U-Net architectures with ResNet blocks and self-attention layers to represent $\epsilon_\theta(x_t, t)$.
- Time is encoded using sinusoidal positional embeddings or random Fourier features.



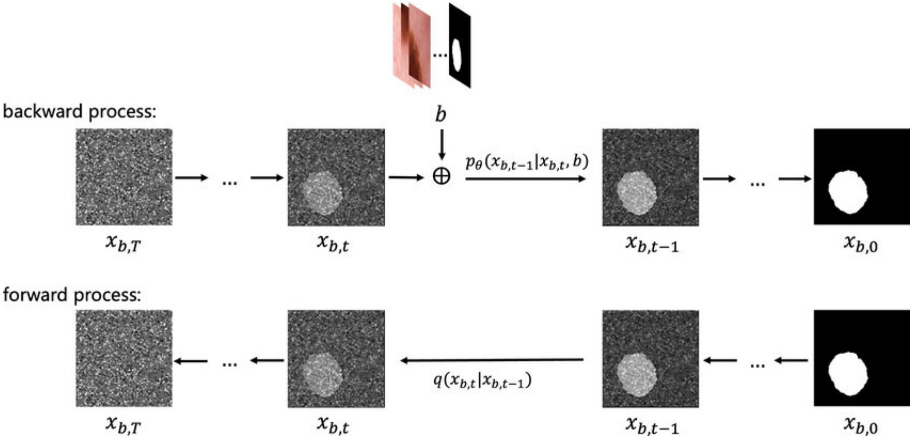


Take **part of the predicted noise** and subtract it from the input. Subtraction amount is defined by the schedule. The first step subtracts only a small part of the predicted noise, later steps subtract more and more of the predicted noise.

Output value

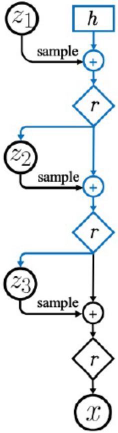
$$x_{t-1} = \frac{1}{\sqrt{a_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}}}\epsilon_\theta(x_t, t)\right) + \sqrt{\beta_t}\epsilon$$

Network Architectures (cont.)



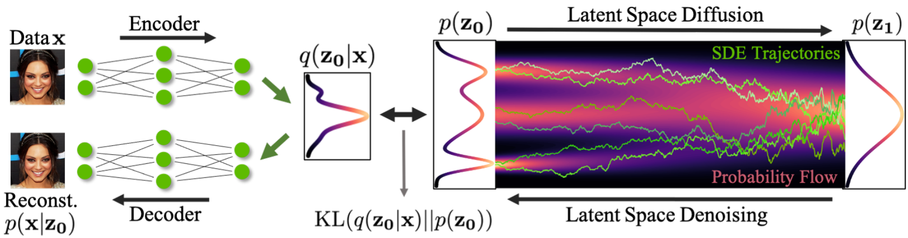
Connection to VAEs

- Diffusion models can be viewed as a special form of hierarchical VAEs (one VAE after another).
- In diffusion models:
  - The encoder is fixed.
  - The latent variables have the same dimension as the data.
  - The denoising model is shared across different timesteps.
  - The model is trained with a reweighted variational bound.



Diffusion Models: **Latent Diffusion Models (LDMs)**

What is a Latent Space?



- A **latent space** is an abstract, compressed space where high-dimensional data (like images) is represented in a lower-dimensional form.
- Think of it as a hidden layer of meaning—each point in this space represents a potential image or concept.
- In classical models like GANs or VAEs, the latent space is directly sampled from (e.g., a vector $\mathbf{z}$), and the generator turns it into an image.
- **Latent** space = "Hidden space of ideas or features"

Where is the Latent Space in Diffusion?

Two main ways latent spaces are used in diffusion:

- **1. Latent Diffusion Models (LDMs):**
  - Diffusion is performed in a compressed latent space, not directly on high-dimensional images.
  - **Pipeline:** Compress image $\rightarrow$ Denoise in latent space $\rightarrow$ Decode back to image.
  - Enables faster, cheaper training and inference, and allows for semantic control (e.g., text prompts, masks).
- **2. Latent Representations in Noise Vectors:**
  - Even in standard diffusion models, the initial noise vector acts as a latent code.
  - Changing the noise vector generates different images or morphs between them.

- **Traditional pixel-space diffusion models are computationally expensive**: they operate directly on high-dimensional images, making both training and inference costly.
- **Latent diffusion reduces computational cost** by performing the diffusion process in a compressed, lower-dimensional latent space—while still maintaining high-quality image generation.
- This efficiency enables more complex and practical applications, such as text-guided generation and flexible image editing.
- **How is this achieved?** Instead of working in pixel space, we learn a compressed latent space using a Variational Autoencoder (VAE) with very low KL divergence weight (e.g., 10^{-6}). This yields a compact representation suitable for diffusion.
- This approach mirrors ideas from autoregressive models (e.g., VQGAN), enabling even more efficient and controllable generation pipelines.

- Text-to-image generation (e.g., Stable Diffusion) uses text embeddings in latent space.
- Image editing by modifying specific parts of the latent.
- Inversion: map a real image to latent, edit, then decode back.

Latent Diffusion Models (Stable Diffusion): Key Concepts

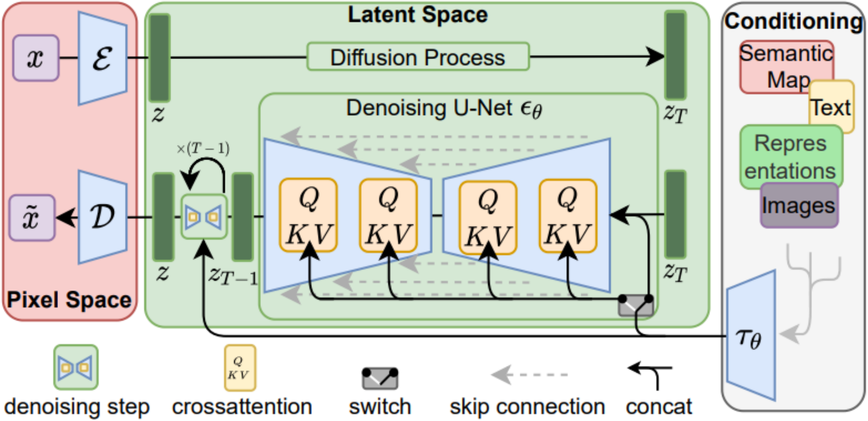
- **Latent Diffusion Models (LDMs):** Operate in a compressed latent space for efficiency.
- **Diffusion Process:** Gradually adds noise to data, then learns to reverse this process.
- **Text Conditioning:** Uses CLIP text encoders to guide image generation from prompts.

Latent Diffusion Models: Architecture Overview

Architecture Overview

- **Encoder:** Maps images to latent space.
- **UNet:** Core denoising network operating in latent space.
- **Decoder:** Reconstructs images from latent representations.
- **Text Encoder:** CLIP-based, encodes prompts for conditioning.

Latent Diffusion Models: Visual Overview



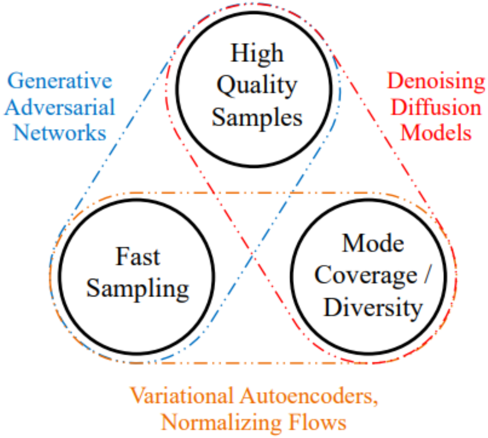
- Initialize from Stable Diffusion.
- Add temporal layers (e.g., 1D Conv, temporal attention).
- Finetune on video data.
- Data pipeline:
  - Scrape **unlabeled** videos from the internet.
  - Use image and video captioners to generate synthetic text labels.
  - Apply aggressive data filtering using several heuristics (CLIP score, optical flow score, aesthetic scores, OCR, etc.).

Diffusion Models: **Issues with Diffusion Models**

Issues with Diffusion Models

- **Slow Sampling:** Diffusion models require hundreds to thousands of iterative steps to generate a sample, making them computationally expensive and slow compared to other generative models.

Trilemma of Generative Learning



Generative Learning Trilemma: Balancing sample quality, diversity, and fast sampling.

[0] Tackling the Generative Learning Trilemma with Denoising Diffusion GANs

Diffusion Models: Current Research Directions

- **Faster Sampling:**
  - Denoising Diffusion Implicit Models (DDIM): Propose a non-Markovian process for faster and deterministic sampling by skipping steps and refining the sample.
- **Improved Training:**
  - Improved Denoising Diffusion Probabilistic Models: Learn the noise variance σ^2 during training, rather than fixing it, for better flexibility and performance.
- **Score-Based Modeling:**
  - Score-Based Generative Modeling through SDEs: Model the gradient of the log-density (score function) using stochastic differential equations for improved sample quality.

Diffusion Models: Current Research Directions (cont.)

- **Combining GANs and Diffusion:**
  - Denoising Diffusion GANs: Use GANs to model each denoising step, addressing the trilemma by improving speed and sample quality.
- **High-Fidelity Generation:**
  - Cascaded Diffusion Models: Employ a cascade of diffusion models at different resolutions to boost sample fidelity.

Diffusion Models: **Applications**

"a man wearing a white hat"

Image Inpainting with GLIDE

[0] GLIDE: Towards Photorealistic Image Generation and Editing with Text-Guided Diffusion Models

Application: DALL.E 2 - OpenAI

a shiba inu wearing a beret and black turtleneck

a close up of a handpalm with leaves growing from it

Text to image generation eith DAll.E 2

0 Hierarchical Text-Conditional Image Generation with CLIP Latents

Fix the CLIP embedding $\mathbf{z}$.

Decode using different decoder latents $\mathbf{x}_T$.

Image Variations

[0] Hierarchical Text-Conditional Image Generation with CLIP Latents

Interpolate image CLIP embeddings $\mathbf{z}_i$

Use different $\mathbf{x}_T$ to get different interpolation trajectories.

Image interpolation

[0] Hierarchical Text-Conditional Image Generation with CLIP Latents

a photo of a cat → an anime drawing of a super saiyan cat, artstation

a photo of a victorian house → a photo of a modern house

a photo of an adult lion → a photo of lion cub

Change the image CLIP embedding towards the difference of the text CLIP embeddings of two prompts.

Decoder latent is kept as a constant.

Text Difference Image interpolation

[0] Hierarchical Text-Conditional Image Generation with CLIP Latents

Imagen - Google

A brain riding a rocketship heading towards the moon.

[0] Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding

Imagen - Google

A dragon fruit wearing karate belt in the snow.

[0] Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding

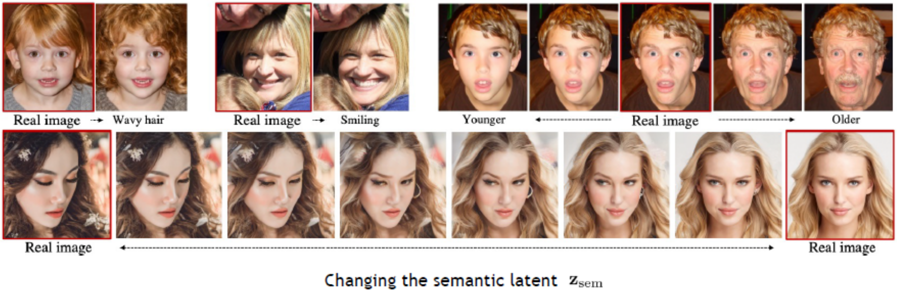
A relaxed garlic with a blindfold reading a newspaper while floating in a pool of tomato soup.

[0] Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding

Application: Nano Banana 2 – Google DeepMind

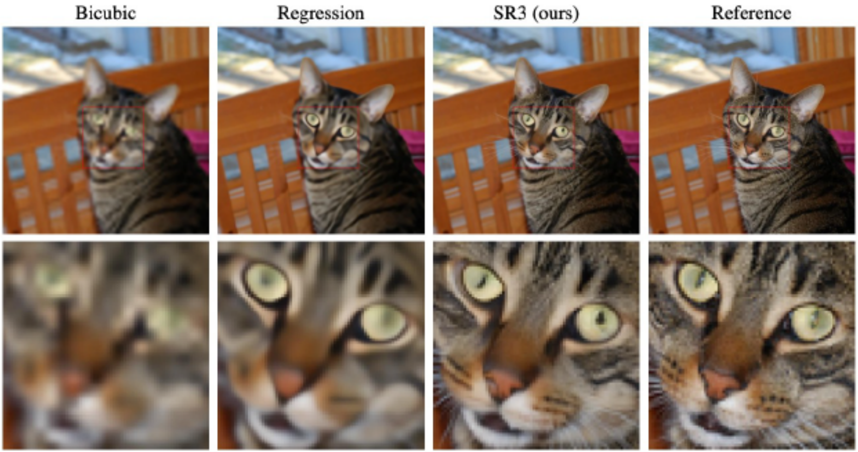
assets/New-Lecture-Stable_Diffusion/nano-banana-2/slide1_world_knowled

Diffusion Autoencoders



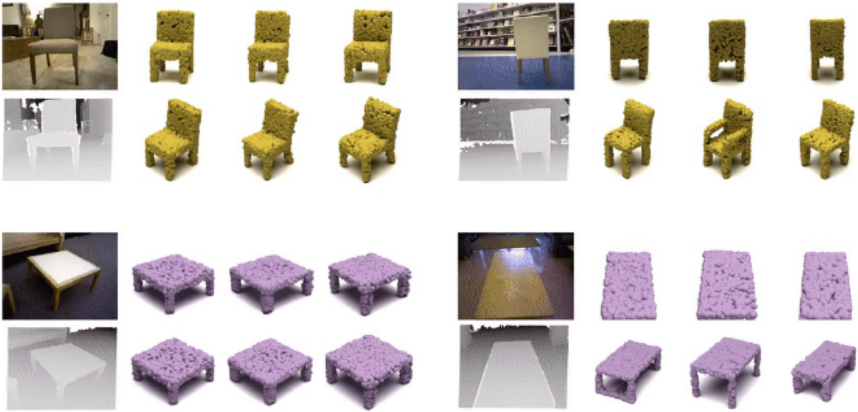
Learning semantic meaningful latent representations in diffusion models

[0] Diffusion Autoencoders: Toward a Meaningful and Decodable Representation



Natural Image Super Resolution $64 \times 64 \longrightarrow 256 \times 256$

[0] Image Super-Resolution via Iterative Refinement



[0] 3D Shape Generation and Completion through Point-Voxel Diffusion

Try It Yourself!

Text to Image generation with stable diffusion.
https://huggingface.co/spaces/stabilityai/stable-diffusion

[0] High-Resolution Image Synthesis with Latent Diffusion Models

Diffusion Models: **Summary & Limitations**

Summary

- Diffusion models add noise to data and learn to reverse it
- Core task: Predict noise at different steps
- Training is efficient; Sampling is iterative and slow
- U-Nets with attention are the winning architecture

Limitations

- **Slow sampling** – iterative process
- **Data-heavy** – needs lots of varied data
- **Distribution miss** – struggles in tails of data
- **Compute needs** – high GPU/time investment
- **Misinformation concerns** – used in deepfake creation

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